

Cargyon

Enterprise ERP — Logística, Supply Chain & Governança Corporativa

38

MÓDULOS

70

MIGRATIONS

378

TABELAS

265

FUNÇÕES/RPC

45

MOTORES IA

8

ÁREAS

Visão geral

Stack

- 🗄 Supabase (PostgreSQL)
- ⚙ Next.js 14 (App Router) + TypeScript + Tailwind
- 🔑 Supabase Auth (JWT) + RLS + RBAC
- 💎 LAIOS (multiagente, insights auto-descobertos, pg_cron 24/7)
- 📱 PWA instalável + offline (service worker)

Arquitetura

- 🏢 Multi-tenant: tenant → company → branch → membership
- 🛡 RLS: habilitado em 100% das tabelas de negócio
- 👤 RBAC: `app.has_permission('<recurso>.<ação>', company_id)`
- 📝 Auditoria: `app.tg_write_audit()` em toda tabela (quem/quando/antes/depois)
- ♻ Soft delete: `deleted_at` / `reason_deleted` (sem DELETE físico)

RLS 100% Backend único Supabase (projeto `ittbmqnwomhkxnv1qjv1`), cérebro LAIOS roda 45 motores de insight auto-descobertos via `pg_cron` a cada 15 min.

Módulos por área (38)

Visão Geral & Inteligência 8

🕒	Cockpit Executivo KPIs logísticos cross-módulo em tempo real	/dashboard
🎯	Command Center (tempo real) Mission control: estado vivo da operação, central de alertas, sala de crise	/comando
⚙	Logistics Control Tower Eventos, SLA, alertas e exceções da cadeia logística	/control-tower
💎	LAIOS — Logistics AI OS Cérebro operacional: multiagentes (WMS/TMS/YMS/Correios), IA preditiva/prescritiva, memória corporativa	/ia-central

	LOGIA (Insights) Insights, previsões e planos de ação logísticos	/logia
	Logistics Data Platform (BI) Data lake/warehouse logístico, catálogo de KPIs, forecast, governança de dados	/analytics
	Auditoria Logística & Custos Auditoria de fretes/operacional, custos, rentabilidade, riscos, compliance logístico, score	/auditoria
	Logistics Planning & Rede Digital twin, mapa de demanda, IA de localização de CD, simulações, ROI/payback	/engenharia-logistica
Estoque & Armazém 6		
	Cadastro Mestre (SKU) SKU, dimensões, peso, categorias, fornecedores, embalagens	/produtos
	Estoque Inteligente Saldos, curva ABC, ponto de pedido	/estoque
	WMS / Armazém Endereçamento, tarefas, ondas, packing	/wms
	WMS Enterprise (IA) Slotting IA, putaway, reabastecimento, RFID, robótica, produtividade, ESG	/operacao-armazem
	Inventário & Rastreio Contagens cíclicas e genealogia de lote	/inventario
	Ativos Retornáveis (RAMS) Pallets/containers/gaiolas/IBCs: empréstimos, retenção, manutenção, ESG	/ativos-retornaveis
Suprimentos 3		
	Suprimentos / Compras Requisição → cotação → pedido → recebimento	/compras
	Demand Planning Histórico, previsões e S&OP	/demanda
	EAM / Manutenção de Frota & Ativos Ativos, ordens de serviço, preventiva, MTTR/MTBF	/manutencao
Expedição & Distribuição 5		
	Expedição Pedidos, picking, packing e embarque	/expedicao
	Smart Shipping Center (IA) Wave picking, escolha de transportadora, embalagem ótima, docas, cargas, gargalos	/central-expedicao
	Distribuição & Last Mile Transferências, cross-dock, hubs, entregas	/distribuicao
	RMA & Logística Reversa Devoluções, conferência, quarentena, retrabalho, reintegração ao estoque	/devolucoes
	Torre de Postagens Etiqueta→PLP→postagem→1ª movimentação, objetos parados, CEP inválido, SLA, alertas	/postagens
Transporte & Pátio 6		
	TMS / Transporte Transportadoras, fretes, rotas e tracking	/tms
	TMS Enterprise / Frota Viagens, custos, combustível, telemetria, manutenção, leilão de fretes, carbono	/frota
	Torre de Transporte Monitoramento em trânsito, ETA, score de risco, ocorrências, heat map, OTIF	/transporte
	Correios (Gestão Enterprise) Contratos, cartões de postagem, PLP, objetos, SRO, auditoria de fretes, SLA	/correios

	YMS / Pátio & Docas Docas, agendamento e pátio	/yms
	YMS Enterprise (Pátio) Portaria/OCR, balanças, filas, carga/descarga, containers, lacres, AI dock scheduler	/patio

Comércio Exterior 1

	Global Trade Management (GTM) Importação/exportação, Incoterms, portos/aeroportos, aduana, containers, drawback, RECOF	/comex
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Cliente & Pós-Venda 2

	Portal do Cliente Logístico Timeline, tracking, ETA, notificações, comprovantes, self-service, devoluções, NPS	/portal-cliente
	Pós-Venda & Ocorrências (CLX) Rastreio público, ocorrências, chamados/SLA, NPS/CSAT	/pos-venda

Plataforma 7

	BPM & Workflows Motor de processos: aprovações multinível, regras de decisão (DMN), SLA, eventos	/processos
	Documentos (ECM / GED) Repositório, versionamento, check-in/out, assinaturas eletrônicas, retenção, busca	/documentos
	Governança de Dados (MDM) Fonte única da verdade: qualidade de dados, deduplicação, linhagem, glossário	/mdm
	Integrações (iPaaS) API Gateway, conectores, event bus, webhooks, fila com retry/DLQ, ETL, chaves de API	/integracoes
	Identidade & Segurança (IAM) Zero Trust, MFA, sessões, PAM, detecção de ameaças, incidentes, certificação de acessos	/seguranca
	Administração da Plataforma Config center (rollback), feature flags, multimoeda, multilíngue, módulos, licenças	/admin
	Super App & Dispositivos PWA instalável, dispositivos, sync offline, push, modos operacionais (coletor/motorista)	/dispositivos

Migrations com schema (67 de 70)

ARQUIVO	TAB.	FN.	TABELAS (AMOSTRA)
000001_foundation.sql	9	1	audit_logs, branches, companies, memberships, permissions, profiles...
000002_master_data.sql	11	0	kit_items, product_categories, product_lots, product_packagings, product_serials, products...
000003_wms.sql	9	1	inbound_receipt_items, inbound_receipts, package_items, packages, pick_waves, stock_balances...
000004_tms.sql	8	0	carriers, drivers, freight_rates, routes, shipment_events, shipment_items...
000005_yms.sql	4	0	dock_appointments, docks, yard_visits, yard_zones
000006_purchasing.sql	8	1	purchase_order_items, purchase_orders, purchase_requisition_items, purchase_requisitions, rfq_items, rfqs...
000007_demand_planning.sql	4	1	demand_forecasts, demand_history, demand_plan_lines, demand_plans
000008_mrp_aps.sql	6	1	bills_of_materials, bom_components, mrp_planned_orders, mrp_runs, routing_operations, work_centers
000009_production.sql	3	1	production_consumptions, production_operations, production_orders
000010_smart_inventory.sql	2	3	reorder_suggestions, stock_snapshots
000011_inventory_traceability.sql	3	1	inventory_count_items, inventory_counts, lot_genealogy

ARQUIVO	TAB.	FN.	TABELAS (AMOSTRA)
000012_shipping.sql	3	1	customers, outbound_order_items, outbound_orders
000013_distribution.sql	4	1	deliveries, delivery_attempts, stock_transfer_items, stock_transfers
000014_control_tower.sql	5	1	alerts, logistics_events, logistics_exceptions, sla_breaches, sla_policies
000015_logia.sql	5	1	logia_action_plans, logia_conversations, logia_insights, logia_knowledge, logia_messages
000016_bi_dashboards.sql	3	1	dashboard_widgets, dashboards, saved_reports
000017_mdm_expansion.sql	7	2	product_brands, product_costs, product_customers, product_documents, product_media, product_suppliers...
000018_mdm_dedup.sql	0	1	
000020_mes.sql	4	1	equipment, process_readings, production_appointments, production_downtimes
000021_manufacturing.sql	2	1	bom_revisions, production_lines
000022_lims.sql	6	1	lab_instruments, lab_methods, lab_reagents, lab_samples, lab_tests, product_specifications
000023_qms.sql	19	2	audit_findings, batch_releases, capas, certificates_of_analysis, complaints, inspection_plan_items...
000024_qms_automation.sql	2	5	electronic_signatures, quality_evidences
000025_eam.sql	6	3	asset_failures, asset_readings, assets, maintenance_plans, wo_parts, work_orders
000026_finance.sql	5	4	bank_accounts, bank_transactions, cost_centers, payables, receivables
000027_finance_efp.sql	10	8	allocation_rules, allocation_targets, bank_statement_lines, bank_statements, customer_credit, dunning_rules...
000028_controlling.sql	7	6	cost_drivers, cost_entries, cost_objects, cost_simulations, dre_accounts, profit_centers...
000029_rma.sql	5	3	return_reasons, rma_events, rma_inspections, rma_items, rma_requests
000030_dispatch_tower.sql	3	4	dispatch_events, dispatch_issues, dispatches
000031_transport_tower.sql	1	4	transport_occurrences
000032_correios.sql	6	4	freight_divergences, plps, postal_contracts, postal_events, postal_objects, postal_services
000033_lct.sql	2	4	incidents, operational_scores
000034_lais.sql	3	5	logistics_audit_findings, logistics_risks, savings_opportunities
000035_ssc.sql	3	5	packaging_boxes, shipping_load_items, shipping_loads
000036_clx.sql	3	4	customer_notifications, customer_occurrences, satisfaction_surveys
000037_tms_enterprise.sql	7	5	fleet_maintenance, freight_contracts, freight_quote_bids, freight_quote_requests, fuel_logs, trip_expenses...
000038_lpnd.sql	1	6	network_scenarios
000039_wms_enterprise.sql	3	7	labor_shifts, slotting_recommendations, warehouse_utilities
000040_yms_enterprise.sql	5	4	containers, gate_events, loading_operations, seals, weighings
000041_rams.sql	4	5	asset_charges, asset_loans, asset_maintenance, returnable_asset_types
000042_gtm.sql	6	3	drawback_acts, hs_classifications, trade_documents, trade_locations, trade_partners, trade_processes
000043_laios.sql	7	4	ai_agents, ai_conversations, ai_decisions, ai_knowledge, ai_messages, ai_runs...
000045_gl.sql	6	9	accounting_periods, chart_of_accounts, gl_reconciliations, journal_entries, journal_entry_lines, posting_rules

ARQUIVO	TAB.	FN.	TABELAS (AMOSTRA)
000046_tax.sql	6	6	fiscal_document_taxes, fiscal_documents, fiscal_obligations, operation_natures, tax_assessments, tax_rules
000047_aa.sql	7	5	asset_categories, asset_insurances, asset_inventory_counts, asset_revaluations, asset_transfers, depreciation_entries...
000048_fpna.sql	5	6	budget_lines, budgets, goals, investment_cases, planning_scenarios
000049_crm.sql	9	5	crm_accounts, crm_activities, crm_campaigns, crm_contacts, crm_leads, crm_opportunities...
000050_oms.sql	4	7	order_events, order_reservations, sales_order_items, sales_orders
000051_cxp.sql	7	9	customer_documents, customer_feedback, exp_rma_requests, knowledge_articles, portal_users, support_tickets...
000052_dcp.sql	9	6	cart_items, carts, cms_pages, marketplace_listings, price_list_items, price_lists...
000053_hcm.sql	12	5	candidates, competencies, departments, employee_benefits, employee_competencies, employees...
000054_pwm.sql	7	6	payroll_event_lines, payroll_items, payroll_runs, terminations, time_bank_movements, time_entries...
000055_bpm.sql	7	5	automation_rules, business_rules, form_definitions, process_definitions, process_events, process_instances...
000056_ecm.sql	6	7	document_comments, document_folders, document_signatures, document_versions, documents, retention_policies
000057_bi.sql	6	5	bi_alerts, bi_dashboards, bi_widgets, data_catalog, kpi_definitions, kpi_snapshots
000058_mdm.sql	6	5	data_lineage, mdm_change_requests, mdm_domains, mdm_duplicates, mdm_glossary, mdm_quality_rules
000059_eip.sql	7	7	api_endpoints, api_keys, event_bus, integration_connectors, integration_flows, integration_messages...
000060_iam.sql	9	7	access_certifications, access_policies, auth_methods, iam_identities, login_attempts, pam_requests...
000061_epa.sql	8	5	config_history, currencies, environments, feature_flags, languages, licenses...
000062_eaaf.sql	6	5	ai_automations, ai_models, ai_prompts, ai_providers, ai_tools, ai_usage_logs
000063_ecc.sql	3	7	command_alerts, crisis_rooms, crisis_updates
000064_esap.sql	5	6	devices, home_widgets, mobile_profiles, push_notifications, sync_queue
000065_elcp.sql	8	5	app_templates, custom_apps, custom_components, custom_entities, custom_forms, custom_pages...
000066_edop.sql	8	8	ops_alerts, ops_incidents, pipeline_runs, pipelines, platform_deployments, service_metrics...
000067_grc.sql	8	5	action_plans, compliance_requirements, continuity_plans, grc_audits, grc_policies, grc_risks...
000068_grc_complement.sql	5	8	authority_delegations, compliance_obligations, governance_bodies, grc_evidence, key_risk_indicators
000069_fix_mdm_dashboard.sql	0	2	